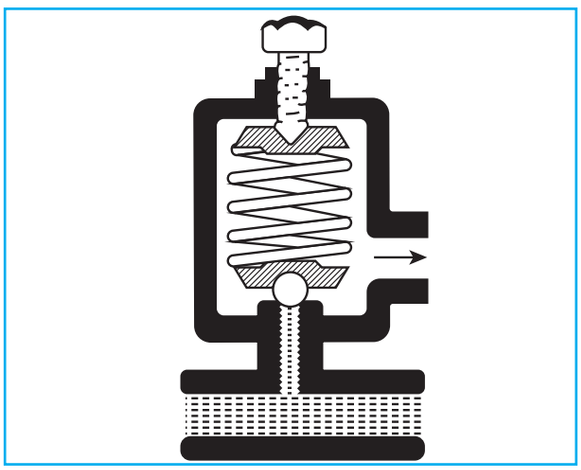


Four way valve

If the plunger reverses, P connects with B and E with A. This four way valve is used for operate and control a double acting hydraulic cylinder.

Pressure Relief valve

The pressure of flow of liquid is limited in a hydraulic circuit. If the pressure exceeds the limit, it affects the hydraulic circuit.



Pressure Relief valve

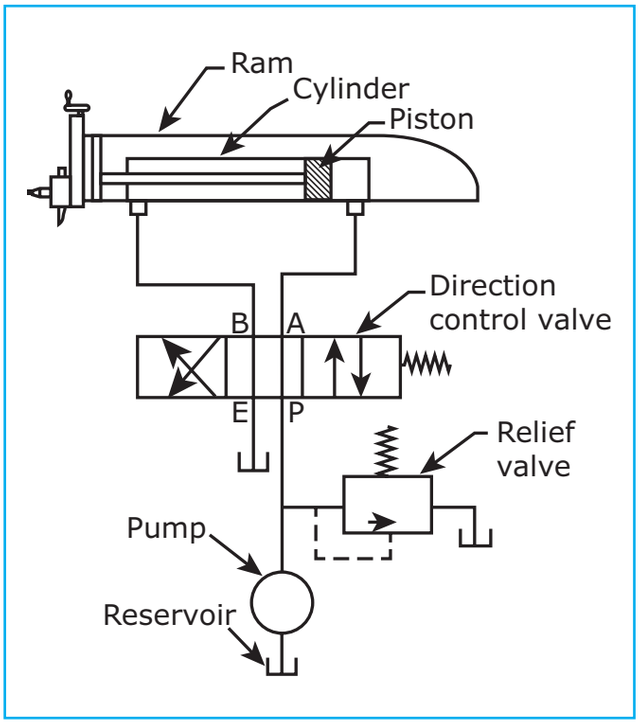
So pressure relief valve is used for control the pressure of the liquid within the required limit.

8.15 Hydraulic circuit of a shaping machine

The important elements of a hydraulic shaper are

1. Constant discharge hydraulic pump
2. Pressure relief valve
3. Direction control valve
4. Cylinder with piston

RLSAAJ



Hydraulic Circuit for Shaping Machine

Working principle

In hydraulic circuit, liquid is pumped by a constant discharge pump into the direction control valve. It has four ways and three stages. If the pressure goes beyond desired limits, the excess pressure is released by the pressure relief valve.

The liquid reaches the right side of the cylinder through the inlet P and outlet A of the valve. Due to the pressure of the liquid, the piston moves towards left.

The tool head moves forward to perform the forward cutting stroke. The liquid on the left side of the piston is sent back to the reservoir through B and E.

Due to the change of position of plunger (spool) inside the direction control valve, the liquid reaches the left side of the cylinder through the inlet P and outlet B. Piston moves towards right. The tool head moves in the opposite direction and this movement is known as the return stroke. The liquid on the

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